Lecture 11: Interest Rate Swaps

Questions

These five multiple-choice questions follow the lecture's reasoning with different numbers. Work through the problem before reading the solutions. Keep full precision until the final answer.

Debt and swap notionals, dates and benchmarks match. Annual rates and semiannual payments are assumed. Net swap cash flow is received minus paid; final borrowing cost is debt payment minus that net.

1. Fix an existing floating liability

A company owes SOFR + 2%. It pays 5% fixed and receives SOFR on a matching swap. What is its final annual borrowing rate?

A. 7% fixed

B. SOFR + 7%

C. 5% fixed

D. SOFR - 3%

2. A payment on the hedge is not failure

The company in Question 1 has $4,000 of debt and swap notional. For a semester with SOFR at 3%, what are its swap net cash flow and final borrowing cost?

A. -$40 net swap cash flow; $60 final cost

B. +$140 net swap cash flow; $0 final cost

C. +$40 net swap cash flow; $60 final cost

D. -$40 net swap cash flow; $140 final cost

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3. Reduce new borrowing cost

A wants fixed: it can borrow at 7% fixed or SOFR + 1%. B wants floating: it can borrow at 9% fixed or SOFR + 4%. A borrows floating; B borrows fixed. A pays B 6.5% fixed and receives SOFR + 1%. What are the final rates?

A. A: 7% fixed; B: SOFR + 4%; neither saves

B. A: 6.5% fixed; B: SOFR + 3.5%; each saves 0.5 percentage points

C. A: 5.5% fixed; B: SOFR + 2.5%; each saves 1.5 percentage points

D. A: SOFR + 7.5%; B: 2.5% fixed

4. Calculate the intermediary's net rate

A bank receives SOFR + 1% from Firm X and 7% fixed from Firm Y. It pays 6% fixed to X and SOFR + 0.5% to Y. Notionals and dates match. What is the bank's annual net swap rate?

A. +1.5%

B. -1.5%

C. SOFR + 2%

D. +0.5%

5. Classify final exposure

Company M borrows at 8% fixed, pays SOFR + 1% in its swap and receives 7% fixed. Company N borrows at SOFR + 2%, pays 6% fixed and receives SOFR + 0.5%. Which pair of final rates is correct?

A. M: SOFR + 2% floating; N: 7.5% fixed

B. M: SOFR + 1% floating; N: 8% fixed

C. M: 2% fixed; N: SOFR + 7.5% floating

D. M: SOFR + 16% floating; N: 4.5% fixed

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Worked solutions

1. Fix an existing floating liability - Answer A

Step 1. Keep the spread. The underlying loan includes a 2% spread that the plain SOFR receipt does not remove.

Step 2. Combine all legs. (SOFR + 2%) + 5% - SOFR = 7%.

Step 3. Identify the exposure. SOFR cancels completely, so the final rate is fixed.

$$(SOFR + 2\%) + 5\% - SOFR = 7\%$$

Interpretation. The locked cost is swap fixed rate plus the loan spread.

Check your reasoning. Do not forget the debt spread when the benchmark cancels.

2. A payment on the hedge is not failure - Answer D

Step 1. Compute the debt payment. 4,000 x (3% + 2%) / 2 = $100.

Step 2. Compute the swap net. Receive 4,000 x 3% / 2 = $60; pay 4,000 x 5% / 2 = $100. Net = -$40.

Step 3. Subtract net cash flow from debt cost. 100 - (-40) = $140, equal to 4,000 x 7% / 2.

$$\text{Cost} = 100 - (-40) = 140$$

Interpretation. The company pays on the swap when the floating loan is cheaper; the combined cost remains fixed.

Check your reasoning. A negative net swap cash flow increases cost when subtracted; do not subtract its absolute value.

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3. Reduce new borrowing cost - Answer B

Step 1. Compute A's preferred fixed rate. A pays SOFR + 1% on debt, pays 6.5% swap and receives SOFR + 1%. The final rate is 6.5%.

Step 2. Compute B's preferred floating rate. B pays 9% debt plus SOFR + 1% swap, less 6.5% received: SOFR + 3.5%.

Step 3. Compare with each direct alternative. A saves 7% - 6.5% = 0.5 points. B saves (SOFR + 4%) - (SOFR + 3.5%) = 0.5 points.

$$A = (SOFR + 1\%) + 6.5\% - (SOFR + 1\%) = 6.5\%; \quad B = 9\% + (SOFR + 1\%) - 6.5\%$$

Interpretation. The correct benchmark is direct borrowing in the market each firm wanted.

Check your reasoning. Do not compare A's final fixed rate with its original floating quote.

4. Calculate the intermediary's net rate - Answer A

Step 1. Add everything received. Receipts are (SOFR + 1%) + 7% = SOFR + 8%.

Step 2. Add everything paid. Payments are 6% + (SOFR + 0.5%) = SOFR + 6.5%.

Step 3. Take received minus paid. The SOFR terms cancel, leaving +1.5% on the matched notional, before expenses and credit losses.

$$[(SOFR + 1\%) + 7\%] - [6\% + (SOFR + 0.5\%)] = 1.5\%$$

Interpretation. This is a net rate, not a dollar amount or return on the bank's equity.

Check your reasoning. Reversing receipts and payments reverses the sign.

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5. Classify final exposure - Answer A

Step 1. Combine M's terms. 8% + (SOFR + 1%) - 7% = SOFR + 2%.

Step 2. Combine N's terms. (SOFR + 2%) + 6% - (SOFR + 0.5%) = 7.5%.

Step 3. Check whether SOFR remains. M retains a SOFR term, so report a floating spread of 2%. N has no SOFR term, so report 7.5% fixed.

$$M = 8\% + (SOFR + 1\%) - 7\%; \qquad N = (SOFR + 2\%) + 6\% - (SOFR + 0.5\%)$$

Interpretation. Identify the algebraic exposure before substituting a particular SOFR scenario.

Check your reasoning. Parentheses matter: subtract the whole receipt, including its spread.